KNOWZONE AI

Product Requirements Document (PRD)

Product vision, user experience, feature requirements, technical architecture, monetization and 12-month growth strategy

Version 1.0 | October 2026

The vision: Turn every student into a smarter, more confident learner.

Knowzone AI is envisioned as an AI-powered learning platform that transforms course materials and educational videos into short explainers, summaries, quizzes and flashcards, while making learning engaging through games, rewards, social study and personalized revision.

Prepared as a product planning document. Pricing, growth and valuation figures are illustrative assumptions to validate—not promises or forecasts.

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1. Executive summary

Knowzone AI should be more than an AI summarizer. It should be a place where students discover knowledge, transform difficult lessons into short videos, master subjects through interactive learning, compete with friends, and earn rewards for real academic progress.

The product combines AI tutoring, short-form educational video, gamified learning and a personalized study companion.

Business ambition: build a company that could justify a $5 million valuation within 12 months. This is an ambitious target, not a guaranteed outcome. A credible case requires retention, paying customers, measurable learning outcomes and a scalable business model.

Initial target market

- **Primary users:** Students preparing for exams, learning difficult subjects or trying to study more efficiently.
- **Secondary users:** Teachers, tutors, parents, schools and educational content creators.
- **Core promise:** Understand more in less time, remember it for longer and make learning enjoyable.
- **Platforms:** Android and iOS mobile apps plus a responsive web application.
- **Revenue:** Freemium subscriptions, school licenses, premium AI usage and selected institutional partnerships.

2. Product look and feel

Visual identity

- **Brand personality:** Smart, energetic, friendly, ambitious and trustworthy.
- **Palette:** Dark navy, electric purple, teal, warm gold and clean white.
- **Typography:** A clean sans-serif such as Inter or Manrope.
- **Interface:** Rounded cards, strong visual hierarchy, short labels and clear progress indicators.
- **Animations:** Small celebrations for completed lessons, badges and level-ups.
- **Accessibility:** Readable text, captions, adjustable playback speed, keyboard support and screen-reader labels.
- **Design principle:** Make learning simple first and entertaining second; avoid cluttering every screen with rewards and pop-ups.

Mobile app navigation

Destination	Purpose
Home	Personalized learning hub, daily goals, continue learning and recommendations.
AI Studio	Create summaries, explainers, flashcards, quizzes and mind maps.
Discover	Explore lessons and educational short videos.
Play	Educational games, themed worlds and missions.
Profile	Progress, rewards, account settings, subscription and referrals.

On desktop, use a left sidebar, central learning workspace and right panel for the AI tutor, daily goals or progress. Add dedicated teacher and school dashboards.

3. Core screens

Screen 1 — Home dashboard

- Greeting and student avatar.
- Daily learning goal and streak.
- Continue where you left off.
- Recommended five-minute lessons.
- Upcoming exams and study reminders.
- Coins, XP and current level.
- Prominent “Create with AI” action.

The first screen should answer: “What should I do next?”

Screen 2 — AI Studio

Students can paste text, upload a PDF, photograph notes or submit an accessible video link. They choose summary depth, explainer length, output type, difficulty and preferred narration language/style. Results are saved to the student’s library.

Screen 3 — Learn feed

- Subject and difficulty tags.
- Captions and narrated explanations.
- Save, replay and share.
- Ask AI about this lesson.
- Quick comprehension question after each video.
- Source link to original material when applicable.

Personalize recommendations around learning goals rather than maximizing endless scrolling.

Screen 4 — Knowzone World

A world of themed learning missions such as Space Lab, Art City, Nature Quest, History Realm and Code Planet. Each world contains interactive challenges, subject missions, boss-level quizzes, collectibles and achievements.

Screen 5 — Progress and rewards

- XP, levels, coins and badges.
- Subject mastery by topic.
- Weekly learning report.
- Personal bests and completed missions.
- Referral progress and earned credits.
- Subscription and usage limits.

Students should see tangible evidence that they are improving.

4. Product requirements and feature specifications

A. AI learning engine — flagship feature

Workflow: input source material → extract and understand content → generate learning assets → measure mastery.

Functional requirements

- FR-01: Upload supported documents, paste text and submit supported video links.
- FR-02: Extract readable text and, where possible, timestamps and source references.
- FR-03: Generate concise, structured summaries with key concepts, definitions and examples.
- FR-04: Generate storyboards, narration and visual assets for short educational videos.
- FR-05: Produce flashcards, multiple-choice questions and practice exercises from the same source.
- FR-06: Answer follow-up questions grounded in uploaded material.
- FR-07: Edit, regenerate, save, export and organize generated content.
- FR-08: Flag uncertain information and provide citations/source references where available.
- FR-09: Display processing status, estimated completion time and usage cost/credit requirements before expensive generation.
- FR-10: Track quiz performance and recommend revision.

Video-generation strategy

Do not generate a fully animated AI video every time a student taps a topic. Start with a lower-cost format using illustrated scenes, motion graphics, text overlays, diagrams and generated or licensed narration. Add richer AI-generated video when it provides measurable value.

For external video platforms, use permitted links, transcripts and APIs. Do not assume every video can be downloaded, transcribed or republished. Respect copyright, licensing and platform terms; link back to original materials.

B. Knowzone AI Tutor

- Explain concepts in simple language and give everyday examples.
- Answer questions about uploaded material.
- Offer hints before revealing answers and explain mistakes.
- Switch between beginner and advanced explanations.
- Support diagrams, equations and step-by-step problem solving where appropriate.
- Recommend practice after an explanation.
- **Teach Me Back:** Ask students to explain a concept in their own words, identify missing ideas and suggest corrections.

C. Smart revision and exam preparation

- Personalized revision calendar and spaced-repetition flashcards.
- Timed mock exams and topic-based practice questions.
- Weak-topic detection and daily/weekly study plans.
- Past-question practice where content is legally available.
- Exam countdown and readiness estimates.
- Revision reminders adapted to the student’s schedule.

Explain how exam-readiness estimates are calculated; do not promise specific exam results.

D. Educational game — Knowzone World

World	Example missions
Space Lab	Planets, gravity and the solar system.
Nature Quest	Identify species and learn ecosystems.
Art City	Colors, artists, design and creativity.
History Realm	Historical mysteries and ordering events.
Code Planet	Logic puzzles and beginner coding.
Math Mountain	Arithmetic, algebra and geometry.
Language Island	Vocabulary, grammar, reading and word puzzles.

Each mission should have a short introduction, an interactive challenge, feedback, a reward and a chance to revisit mistakes.

Rewards system

- XP measures progress and determines level.
- Coins purchase cosmetic items, avatars, themes and other non-essential game items.
- Badges recognize accomplishments.
- Streaks encourage regular study without harshly punishing missed days.
- Seasonal challenges refresh the experience; leaderboards can compare friends or classrooms with privacy controls.

Keep coins separate from cash and subscription credits. Avoid cash-out rewards and gambling-like mechanics. Reward learning, not spending.

E. Referral and credits

- Each student receives a unique referral link/code.
- A friend signs up and completes a qualifying learning activity.
- The referral is validated; both receive a disclosed reward.
- The inviter can track pending, confirmed and rejected referrals.
- Start with modest AI-generation credits, monthly caps and anti-abuse controls.

Do not reward unverified registrations. Detect duplicate accounts, referral loops, automated sign-ups and suspicious device activity. Publish reward rules before launch.

F. Social learning

- Private study groups.
- Shared notes and flashcard decks.
- Friend-versus-friend quiz battles.
- Group exam challenges and class leaderboards.
- Shared revision schedules.
- Teacher-created assignments.
- Safe, moderated discussion spaces.

Start with study groups and quiz challenges. Open public messaging and user-generated feeds add moderation, privacy and child-safety risks, so they should come later.

G. Differentiating features

- **Personal Learning DNA:** Learn which concepts are mastered and which need revision; personalize without permanently labeling students.
- **Voice Tutor:** Spoken questions, audio explanations and verbal recall practice.
- **Low-data/offline learning:** Download summaries, flashcards and videos over Wi-Fi; sync later.
- **Snap-to-Study:** Photograph notes or textbook excerpts and turn them into structured revision material, subject to quality and copyright limits.
- **Knowzone for Schools:** Teacher assignments, class-level mastery reports and school licenses.
- **Future Skills Academy:** Coding, AI literacy, design, entrepreneurship, financial literacy and communication; add certificates only with credible assessment standards.

Recommended core: AI Studio + smart revision + Knowzone World. Offline access and school tools are the next expansion. Voice, advanced social features and career certificates can follow after demand is proven.

5. Web application requirements

Web area	Required functionality
Student dashboard	Courses, study plan, streaks, saved content and upcoming exams.
AI workspace	Upload files, compare summaries, edit notes and generate study resources.
Video workspace	Explainers, transcripts, timestamps and quizzes.
Study library	Search, filter, tag, organize and export materials.
Game center	Worlds, missions and achievements.
Teacher dashboard	Create assignments, manage classes and monitor learning progress.
School administration	Manage seats, permissions, billing and institution-wide usage.
Admin console	Manage users, content, reports, subscriptions, AI costs and abuse.

Support keyboard navigation, responsive layouts, autosaving, resumable uploads and synchronized learning progress between mobile and web.

6. User journeys

Journey A — Turn a difficult lesson into an easy one

- Onboarding: choose education level, subjects, exam goals and learning preferences.
- Upload a biology PDF or paste a supported educational video link.
- Knowzone generates a concise summary, explainer video and short quiz.
- Student watches and answers five questions.
- The app identifies weak concepts and recommends flashcard review.
- Student earns XP, saves the material and may receive a revision reminder.

Aim for meaningful learning value within the first few minutes, without a long onboarding process.

Journey B — Invite a friend

- Open Profile → Invite Friends.
- Share a referral link through a supported messaging app.

- Friend registers and completes a qualifying activity.
- System validates the referral and awards credits.
- Both students use credits on eligible AI features, within published limits.

Journey C — School adoption

A teacher creates a class, assigns a topic, sets a deadline and receives a class-level report showing common knowledge gaps. Students receive personalized practice. The school pays for managed access, student seats, reporting and administrative controls.

7. Monetization and pricing

Use freemium pricing with a low entry point and several revenue options. Prices below are proposed experiments—not established market rates—and should be validated with students and parents.

Plan	Proposed price	Included value
Free	■0	Limited summaries/quizzes, selected short lessons, basic flashcards, daily missions, basic progress and limited AI tutor usage.
Knowzone Plus	■2,500–■5,000/month	Higher AI limits, more explainers, advanced quizzes/revision, offline downloads, personalized plans and richer insights.
Knowzone Pro	■6,000–■10,000/month	Higher generation limits subject to fair use, advanced exam practice, larger libraries, enhanced tools and priority processing where feasible.
Schools/institutions	Custom	Annual or per-student licenses, teacher dashboards, assignments, reports, bulk licensing and support.

Other revenue opportunities

- Premium credit packs for additional AI video/content generation.
- Family plans with multiple student accounts.
- Creator marketplace for licensed educational packs, with a disclosed commission.
- School and tutoring partnerships.
- Institutional content partnerships and licensed curricula.

Consider lower-cost weekly or term-based packages if research supports them. Avoid intrusive ads in lessons, selling student data or monetizing children’s personal information.

8. Security, quality and performance

Area	Requirement
Accounts	Secure authentication, account recovery and optional social login.
Data protection	Encryption in transit/at rest, access controls and defined retention periods.
Child safety	Age-appropriate experiences, parental consent where required, reporting and moderation.
AI quality	Source grounding, factual checks, uncertainty indicators and user feedback.
Copyright	Authorized uploads, permitted transcript processing, takedown procedures and licensing controls.
Reliability	Background jobs, retries, processing status and recovery from failed generations.
Performance	Fast loading, resumable uploads and responsive interfaces.
Low connectivity	Offline access to eligible saved content and reliable synchronization.
Accessibility	Captions, readable contrast, keyboard navigation and assistive technology support.
Cost controls	Per-user quotas, model routing, caching, abuse detection and generation budgets.
Payments	Transparent pricing, renewal terms, subscription management and refunds.
Privacy	Minimal collection, deletion/export workflows and appropriate analytics consent.

Assess applicable Nigerian data protection requirements and any laws governing minors and international users. Have qualified legal counsel review the product before launch.

9. Recommended technology architecture

A modular monolith plus background workers is a sensible starting point. Avoid an expensive microservices architecture before scale requires it.

Suggested stack

- **Mobile:** React Native with Expo.
- **Web:** Next.js with TypeScript.
- **Backend:** NestJS or a well-structured TypeScript backend; Python services for specialized AI pipelines if needed.
- **Database:** PostgreSQL, with pgvector where semantic search is useful.
- **File storage:** S3-compatible object storage.
- **Background jobs:** Redis-backed queues or a managed job service.
- **AI:** Provider abstraction so models can be selected for cost and quality across summarization, tutoring, speech and media generation.
- **Video rendering:** FFmpeg-based rendering with reusable templates and asynchronous workers.
- **Payments:** Providers supporting target markets, currencies and app-store requirements.
- **Analytics:** Product funnels, subscriptions, learning outcomes and server-side cost monitoring.

Log usage per user and feature, and keep AI providers interchangeable.

10. MVP scope and releases

Release 1 — Validate the core (Months 1–3)

- Accounts and onboarding.
- PDF, text and supported-link input.
- AI summaries and structured notes.
- Template-based narrated explainers.
- Quizzes and flashcards.
- Basic AI tutor grounded in saved content.
- Study library and progress tracking.
- Coins, XP, daily goals and simple missions.
- Subscription/usage-limit infrastructure.
- Basic analytics and feedback.

Release 2 — Build retention (Months 4–6)

- Personalized revision schedules and spaced repetition.
- Referral credits and anti-fraud controls.
- More game worlds and seasonal challenges.
- Offline notes, flashcards and eligible videos.
- Study groups and friend quiz challenges.
- Improved curriculum-aligned content.
- Early school and tutoring pilots.

Release 3 — Scale distribution (Months 7–12)

- School administration and teacher dashboards.
- Paid institutional licenses.
- Voice tutor and additional languages.
- Advanced adaptive learning.
- Creator partnerships and licensed marketplace.
- Expanded exam-preparation coverage.
- Improved AI video quality and rendering efficiency.
- Regional expansion after product-market fit.

Initial team

- One product-minded technical lead.
- Two full-stack or cross-platform engineers.
- One AI/backend engineer.
- One product designer.
- One learning-content specialist or instructional designer.
- One growth and partnerships lead, potentially part-time initially.

QA, security, legal and content expertise are also needed through employees or contractors. Prioritize the core learning loop before investing heavily in the game world.

11. 12-month business and growth strategy

Period	Main objective	Evidence of progress
Months 1–2	Interview students and prototype	30–50 interviews, usability tests and willingness-to-pay signals.
Month 3	Closed beta	100–300 active testers and measured first-week retention.
Months 4–5	Improve learning and retention	Completion, repeat usage and initial paid conversions.
Month 6	Public launch	Stable onboarding, payments and a repeatable acquisition channel.
Months 7–9	Monetization and partnerships	Paying cohorts, referral economics and school pilots.
Months 10–12	Scale proven channels	Stronger retention, predictable revenue and repeatable growth.

Acquisition channels to test

- Campus ambassadors who demonstrate Knowzone in study groups and university communities.
- Educational creators who demonstrate real lesson transformations.
- Exam-preparation communities with curriculum-aligned tools and practice content.
- Referral loops that reward successful learning-driven invitations.
- School partnerships with teacher dashboards and measurable outcomes.
- Search and social content that provides free useful summaries, quizzes and explainers.
- Study challenges timed around exams and academic calendars.

Track which channels bring users who stay active and pay. Downloads alone do not establish a successful acquisition channel.

12. Path toward a $5 million valuation

A valuation is not cash in the bank. A company valued at $5 million is not necessarily earning $5 million, and the valuation does not guarantee a buyer. Investors consider recurring revenue, growth, retention, margins, defensibility, market size, execution risk and financing conditions.

Illustrative month-12 targets

Metric	Illustrative target
Monthly active users	100,000
Paying individual users	5,000
Revenue per subscriber	■4,000/month
School/institution contracts	10–20 pursued

Consumer subscription calculation: 5,000 × ■4,000 = ■20,000,000 in monthly gross subscription billings, before payment fees, taxes, refunds, AI costs and operating expenses. These are planning targets, not a prediction. Institutional revenue

is additional only once contracts are secured.

Illustrative valuation equation

If, purely for illustration, investors valued the business at 5× annualized recurring revenue:

$5,000,000 ÷ 5 = $1,000,000 annualized recurring revenue.

$1,000,000 ÷ 12 ≈ $83,333 monthly recurring revenue.

The multiple is an assumption, not a guarantee or universal industry standard. The actual valuation could be substantially higher or lower.

Metrics investors may care about

- **Retention:** Day-1, day-7 and day-30 retention by cohort. An initial planning target might be 25–35% day-30 retention for a broad consumer learning audience; validate against actual cohorts and comparable products.
- **Learning outcomes:** Quiz improvement, revision-plan completion, recall after several days and student-reported usefulness.
- **Revenue and margins:** Free-to-paid conversion, MRR, subscriber churn, revenue per paying user and contribution margin after AI, media, payment and support costs.
- **Efficient growth:** Customer acquisition cost, referral conversion, organic acquisition, payback period and first meaningful learning-session completion.

The retention range is a proposed internal target, not a verified benchmark for this specific market. Opening the app should not count as meaningful learning.

A useful external reference for further diligence is RevenueCat's 2026 subscription-app research, which reports that AI apps can monetize well but have weaker median long-term retention than non-AI apps. This reinforces the need to prove that learning and revision bring students back. Source: https://www.revenuecat.com/

13. Risks and mitigations

Risk	Mitigation
AI summaries contain errors	Ground outputs in sources, verify questions, expose citations and collect reports.
AI video costs become excessive	Use templates, caching, quotas and tiered generation quality.
Students try once and leave	Build revision schedules, mastery feedback and useful daily goals.
Referral abuse drains budget	Validate qualifying activity, cap rewards and detect suspicious accounts.
Copyright violations	Use licensed content and permitted processing; respect platform terms.
Game mechanics distract	Reward recall and progress rather than screen time.
Slow school adoption	Validate consumer demand first and run small paid institutional pilots.
Student privacy compromised	Minimize collection, restrict access and establish child-safety controls.
Too many features delay launch	Ship the core AI-to-learning loop before advanced social and gaming features.

14. First 90-day validation plan

The most important early question is whether students repeatedly use Knowzone to understand difficult topics and whether some will pay for it.

- Interview 30–50 students across different education levels.
- Test a clickable prototype with at least 10 students.
- Build a working flow from PDF or text to summary, explainer and quiz.
- Run a closed beta with 100–300 testers.
- Measure first-session completion, day-7 retention and day-30 retention.
- Compare learning outcomes with a simple baseline assessment.
- Test willingness to pay at different price points.
- Calculate the cost of each summary, quiz and generated video.
- Run a small referral experiment.
- Decide whether the evidence supports expanding the game and school platform.

Do not scale paid acquisition until users receive value, return and can be monetized at a sustainable cost.

15. Final product recommendation

Position Knowzone AI as **the personal AI learning platform that turns any lesson into an experience students can understand, practice and master.**

Five strategic priorities

- **Own the transformation:** Make the path from difficult source material to understandable notes, short videos and practice questions exceptionally good.
- **Own the learning outcome:** Use quizzes, recall, spaced repetition and adaptive revision to help students master subjects.
- **Make learning rewarding:** Build a distinctive exploration game that makes academic progress enjoyable without undermining learning.
- **Build distribution into the product:** Use referrals, study groups, creator partnerships and school relationships to grow sustainably.
- **Build a real business:** Control AI costs, test willingness to pay, measure retention and create recurring revenue from individuals and institutions.

A successful Knowzone AI would not simply summarize the internet. It would become the place a student goes when confused, when exams are approaching, when they want to learn something new and when they want to see how much they have improved.

RECOMMENDED NEXT STEP

Turn this PRD into a screen-by-screen UI/UX specification and clickable prototype, beginning with Home, AI Studio, the learning player, Knowzone World and Profile. Validate it with students before committing to the full build.